

# Transformada de Fourier

**Definición 1 (Transformada de Fourier).** Sea  $(H, \langle \cdot, \cdot \rangle)$  un espacio prehilbert y  $\{u_n\}_{n \in \mathbb{N}}$  un sistema ortonormal en  $H$ ,  $\mathcal{F}$  es la transformada de Fourier asociada a  $\{u_n\}_{n \in \mathbb{N}}$

$$\Longleftrightarrow \forall x \in H : \mathcal{F}(x) = (\langle x, u_n \rangle)_{n \in \mathbb{N}}.$$

A los escalares  $\hat{x}_n := \langle x, u_n \rangle$  se les llama coeficientes de Fourier de  $x$  respecto a  $\{u_n\}_{n \in \mathbb{N}}$ .

## Referenciado en

- serie-fourier-l1
- obs-sumas-parciales-serie-fourier-convolucion-nucleo-dirichlet
- sumacion-cesaro
- teo-riesz-fischer

### **Temporary page!**

L<sup>A</sup>T<sub>E</sub>X was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L<sup>A</sup>T<sub>E</sub>X now knows how many pages to expect for this document.